

Table 5
Mathematic modeling.

Indices:	
$i = 1, \dots, I$	The producer locations (pistachio orchard)
$j = 1, \dots, J$	The potential processing center
$k = 1, \dots, K$	The potential pistachio factory locations
$e = 1, \dots, E$	The potential oil extraction center locations
$q = 1, \dots, Q$	The potential composting center locations
$s = 1, \dots, S$	The potential cosmetic factories locations
$n_1 = 1, \dots, N_1$	The pistachio market locations
$n_2 = 1, \dots, N_2$	The oil market locations
$n_3 = 1, \dots, N_3$	The cosmetic customer locations
$b = 1, 2$	Type of processing center products
$m_1 = 1, \dots, M_1$	Some producers as compost customers
$m_2 = 1, \dots, M_2$	The compost markets
$m = 1, \dots, M$	Total compost's customers
$M = M_1 + M_2$	
Parameters:	
Fu_j	Fixed cost of processing center j
Fy_q	Fixed cost of composting center q
Fw_k	Fixed cost of pistachio factory k
Fr_e	Fixed cost of oil extraction center e
Fv_s	Fixed cost of cosmetic factory s
Cl_i	Production cost per unit of product from producer i
Cy_q	Production cost per unit of product for composting center q
Cw_k	Roasting and packing cost per unit of product for pistachio factory k
Cr_e	Extracting cost per unit of product for oil extraction center e
Cv_s	Production cost per unit of products for cosmetic factory s
Cub_{ij}	Processing cost per unit of product type b for processing centers j
CX_{ij}	Shipping cost per unit of picked pistachio shipped from the producer (pistachio orchard) i to processing center j
CK_{jk}	Shipping cost per unit of open-mouth pistachio shipped from processing center j to pistachio factory k
CE_{je}	Shipping cost per unit of raw kernel shipped from processing center j to oil extraction center e
CJ_{jq}	Shipping cost per unit of processing waste shipped from processing center j to composting center q
CS_{es}	Shipping cost per unit of products from oil extraction center e to cosmetic factory s
CN_{en_2}	Shipping cost per unit of product shipped from oil extraction center e to oil customer n_2
CQ_{eq}	Shipping cost per unit of products from oil extraction center e to composting center q
Cl_{sn_3}	Shipping cost per unit of product shipped from cosmetic factory s to cosmetic customer n_3
Cp_{kn_1}	Shipping cost per unit of product shipped from pistachio factory k to pistachio customer n_1
Cd_{qm}	Shipping cost per unit of produced compost shipped from composting center q to compost customer m
Cl_i	Production cost per unit of product from producer i
DC_m	The demand for compost by customer m
Dp_{n_1}	The demand of open-mouth pistachio by pistachio customer n_1
Dun_2	The demand for oil by oil customer n_2
Dsn_3	The demand of cosmetic by cosmetic customer n_3
β	Weight loss percentage in drying operations
θ_b	Waste percentage of product type b from processing center j
γq	Production rate of compost from composting center q
γs	Production rate of cosmetic product from cosmetic factory s
γk	Production rate of pistachio factory k
λ	Waste percentage of oil extracting process in oil extraction center e
Cpa_i	Production capacity for producer i
Cpu_j	Production capacity for processing centers j
Cpy_q	Production capacity for composting center q
Cpw_k	Production capacity for pistachio factory k
Cpr_e	Production capacity for oil extraction center e
Cpv_s	Production capacity for cosmetic factory s
Decision variable:	
X_{ij}	Quantity of picked pistachio shipped from producer i to processing center j
GO_{jk}	Quantity of product open-mouth pistachio shipped from processing center j to pistachio factory k
Gr_{jk}	Quantity of raw kernel shipped from processing center j to oil extraction center e
GW_{jk}	Quantity of waste shipped from processing center j to composting center q
O_{en_2}	Quantity of oil shipped from oil extraction center e to oil customer n_2
OC_{es}	Quantity of oil shipped from oil extraction center e to cosmetic factory s
OW_{eq}	Quantity of waste shipped from oil extraction center e to composting center q
L_{sn_3}	Quantity of product shipped from cosmetic factory s to cosmetic customer n_3
P_{kn_1}	Quantity of product shipped from pistachio factory k to pistachio customer n_1
D_{qm}	Quantity of produced compost shipped from composting center q to compost market m
U_j	1 If processing center j is opened, 0 otherwise
Y_q	1 If composting center q is opened, 0 otherwise
W_k	1 If pistachio factory k is opened, 0 otherwise
R_e	1 If oil extraction center e is opened, 0 otherwise
V_s	1 If cosmetic factory s is opened, 0 otherwise

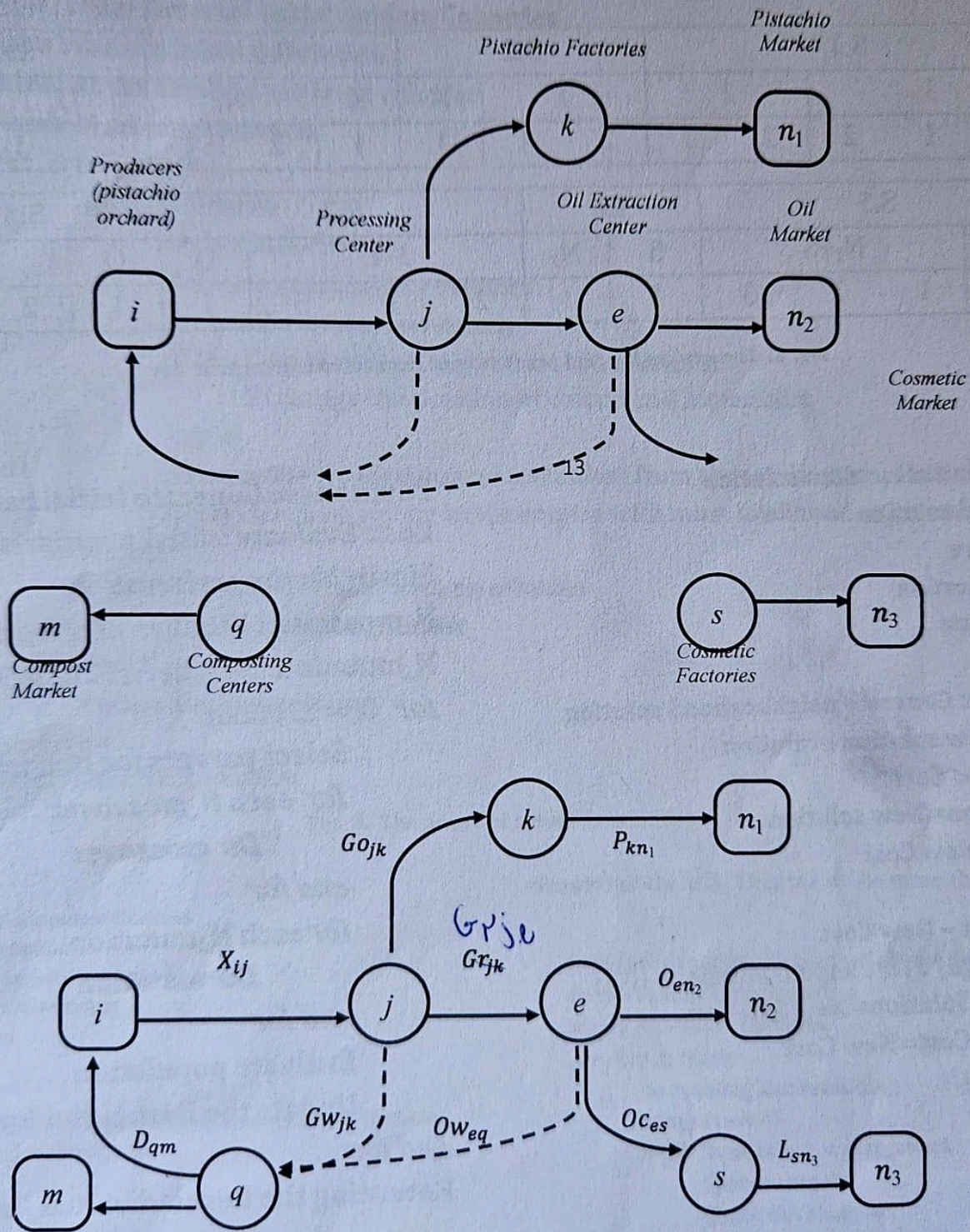


Fig. 4. Proposed Pistachio Closed-loop Supply chain.

Table 7
Model parameters' surface.

Factors	Levels	Factors	Levels
Fu_j	$U(20000, 40000)$	β	$= 0.3$
Fy_q	$U(10000, 20000)$	θ_b	$= [0.4, 0.1, 0.5]$
Fw_k	$U(20000, 50000)$	γq	$= 0.6$
Fre	$U(4000, 8000)$	γs	$= 2$
Fv_s	$U(40000, 80000)$	γk	$= 1.01$
CI_i	$U(80, 120)$	λ	$= 0.4$
Cy_q	$U(50, 100)$	Cpa_i	$U(80000, 100000)$
Cw_k	$U(100, 150)$	$Cpu_j, Cpy_q, Cpw_k, Cpr_e, Cpv_s$	$U(20000, 40000)$
Cr_e	$U(40, 80)$	CI_i	$U(20, 40)$
Cv_s	$U(150, 180)$	DC_m	$U(200, 500)$
Cu_{1j}	$U(60, 100)$	DP_{n1}	$U(200, 400)$
Cu_{2j}	$U(80, 120)$	Du_{n2}	$U(100, 200)$
$CX_{ij}, CK_{jk}, CE_{je}, CJ_{jq}, CS_{es},$ $CN_{en2}, CQ_{eq}, CI_{sn3}, CP_{kn1},$ Cd_{qm}	$U(20, 40)$	Ds_{n3}	$U(20, 80)$

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Procedure 1 : Decoding of the chromosome for transportation tree

Input : K : set of sources, J : set of depots,
 b_j : demand on depot j , $\forall j \in J$,
 a_k : capacity of source k , $\forall k \in K$,
 c_{kj} : transportation cost of one unit of product from source k to depot j ,
 $\forall k \in K, \forall j \in J$,

$v(k+j)$: chromosome, $\forall k \in K, \forall j \in J$,

Output : g_{kj} : the amount of product shipped from source k to depot j

Step 1. $g_{kj} \leftarrow 0$, $\forall k \in K, \forall j \in J$,

Step 2. $l \leftarrow \arg \max \{v(t), t \in K | + | J |\}$; select a node

Step 3. If $l \in K$, then $k^* \leftarrow l$; select a source

$j^* \leftarrow \arg \min \{c_{kj} | v(j) \neq 0, j \in J\}$; select a depot with the lowest cost

else $j^* \leftarrow l$; select a depot

$k^* \leftarrow \arg \min \{c_{kj} | v(j) \neq 0, k \in K\}$; select a source with the lowest cost

Step 4. $g_{k^*j^*} \leftarrow \min \{a_{k^*}, b_{j^*}\}$; assign available amount of units

Update availabilities on source (k^*) and depot (j^*)

$$a_{k^*} = a_{k^*} - g_{k^*j^*} \quad b_{j^*} = b_{j^*} - g_{k^*j^*}$$

Step 3. If $a_{k^*} = 0$ then $v(k^*) = 0$

If $b_{j^*} = 0$ then $v(j^*) = 0$

Step 5. If $v(|K| + |J|) = 0$, $\forall j \in J$, then calculate transportation cost and return, else goto Step 1.

Fig. 3 Decoding algorithm for the priority-based encoding